Tsung-Yi Ma

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Senior AI Engineer with a strong mathematical background (NTU Math; Phi Tau Phi Honor Society). Focused on Generative AI, real-time speech systems, and mathematical optimization. ASUS Global Role Model Award recipient.

Experience

ASUSTEK COMPUTER INC. — New Taipei City, Taiwan

Senior AI Engineer — *Aug. 2025 – Present*

AI Engineer — *Nov. 2023 – Aug. 2025*

- **2025 Global Role Model Award**: Selected for the **Outstanding Contribution Award** (HQ selection, Top $<$1% company-wide) for exceptional performance and high-impact project delivery.
- **AI Interview Agent (GAI Recruiter)**: Architected a real-time streaming speech recognition pipeline using **Python (asyncio), Ray, and WebSockets**. Implemented a **sliding-window approach** to maintain sub-500 ms streaming latency. Optimized inference parameters (VAD, beam size) to significantly reduce hallucinations.
- **Production Scheduling Optimization**: Designed mathematical models (MIP/LP) using **CPLEX** and **SCIP** solvers. Restructured constraints to reduce execution time from **24+ hours to $<$1 hour**. Simulations validated a **10% increase in production output** under raw material shortages compared to traditional FIFO methods.
- **License Plate Voice Recognition**: Fine-tuned **OpenAI Whisper** for specialized alphanumeric recognition, boosting accuracy to $>$**80%**. Implemented on-the-fly data augmentation (A-law encoding, noise injection) to ensure model robustness.

Education

National Taiwan University — Taipei, Taiwan

Master of Mathematics — *Sep. 2021 – Jun. 2023*

- **The Phi Tau Phi Scholastic Honor Society**: Inducted as an Honorary Member, an academic distinction typically awarded to top Master's graduates.
- **CS Foundations**: Data Structures & Algorithms (**A+**), Machine Learning (**A**). *Self-study:* Operating Systems, Computer Networks, Computer Architecture.
- **Thesis**: "Introduction to the Ising Model and the Ising Model with Disorders". Applied rigorous analysis to statistical physics models.

National Taiwan University — Taipei, Taiwan

Bachelor of Mathematics — *Sep. 2017 – Jun. 2021*

- **Research**: Selected for **NCTS Undergraduate Summer Research Program** (Noncommutative Probability).
- **Honors Math**: Completed Honors tracks in Analysis, Algebra, Geometry, and Complex Analysis (Advanced curriculum replacing standard core requirements).

Taipei Municipal Chien Kuo High School — Taipei, Taiwan

***Class of Science** (Honors track for advanced scientific training)* — *Sep. 2014 – Jun. 2017*

Projects

Distributed Whisper Streaming | *Python, Ray, faster-whisper* — GitHub

- Added a **RayQueue**-based routing layer to share GPU-bound Whisper inference workers across concurrent streaming sessions.

SQLite File Reader | *Rust, SQLite storage internals, B-tree pages* — GitHub

- Built a read-only parser for SQLite database files, traversing table/index B-tree pages and decoding varints, record serial types, and schema entries for simple SELECT queries.

Technical Skills

Languages: Python (Primary); project experience with Rust, Go, and TypeScript; basic SQL
AI & Speech: PyTorch, OpenAI Whisper, faster-whisper, VAD, streaming inference
Optimization: Mathematical Programming, MIP/LP, CPLEX, SCIP
Backend & Systems: FastAPI, async Python, WebSockets, Ray, Linux
Developer Tools: Git, Ruff, Pyright